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A new skew integer valued time series process

Marcelo Bourguignon [a,*], Klaus L.P. Vasconcellos [b]

[a] Universidade Federal do Rio Grande do Norte, Departamento de Estatística, Lagoa Nova, 59078-970, Natal, RN, Brazil

[b] Universidade Federal de Pernambuco, Departamento de Estatística, Cidade Universitária, 50740-540, Recife, PE, Brazil

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ABSTRACT

In this paper, we introduce a stationary first-order integer-valued autoregressive process with geometric–Poisson marginals. The new process allows negative values for the series. Several properties of the process are established. The unknown parameters of the model are estimated using the Yule–Walker method and the asymptotic properties of the estimator are considered. Some numerical results of the estimators are presented with a brief discussion. Possible application of the process is discussed through a real data example.

1. Introduction

Integer-valued times series with support in $\mathbb{Z} = \{\ldots, -2, -1, 0, 1, 2, \ldots\}$ are very frequent in practice. The need for modeling and analyzing counting data (negative and positive) occurs in many fields of real life such as medicine [11], image analysis [8], sports applications [12], financial applications [2] and so on.

Models for time series defined on the set $\mathbb{Z}$ of integers have been discussed by various researchers. Kim and Park [13] introduced an integer-valued autoregressive (INAR) process of order p with signed binomial thinning operator. Zhang et al. [23] introduced pth-order integer valued autoregressive processes with a signed generalized power series thinning operator. Kachour and Truquet [9] introduced a more general class, based on a modified version of the generalized thinning operator, also called the signed thinning operator. Kachour and Yao [10] proposed the first-order rounded integer-valued

* Corresponding author.
E-mail addresses: m.p.bourguignon@gmail.com (M. Bourguignon), klaus@de.ufpe.br (K.L.P. Vasconcellos).

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autoregressive process, based on the rounding operator. Some of these models arise as the difference between two discrete distributions. For example, Freeland [6] defined the true integer-valued autoregressive process of order one as the difference of two Poisson INAR(1) processes [1] which requires observing the two processes. Recently, Barreto-Souza and Bourguignon [4] introduced a stationary AR(1) process on $\mathbb{Z}$ with skew discrete Laplace marginals (which are distributed as a difference between two geometric random variables); this model is named skew INAR(1) process on $\mathbb{Z}$.

In a similar manner, we introduce a stationary first-order integer-valued autoregressive process with geometric–Poisson marginals (which are distributed as a difference between geometric and Poisson random variables), named new skew INAR(1) process (NSINAR(1)). Additionally, we will provide a comprehensive account of the mathematical properties of the proposed new process. The motivation for such a process arises from its potential in modeling and analyzing integer valued time series for which the non-negative part presents greater overdispersion than the negative part. The price change data can satisfying this interesting condition (see Section 5). In this context, it is plausible to consider the difference between two distinct processes, one with greater overdispersion for the positive part (which we will model as geometric) and another with lower variability for the negative part (which we will model as Poisson). The new process can be used very well for modeling the change in price of a commodity, stock or other financial product.

The rest of the paper unfolds as follows. In Section 2, the NSINAR(1) process is introduced. In Section 3, some of its properties are outlined. Estimation methods for the model parameters are proposed, while numerical results from Monte Carlo simulation experiments are presented and discussed in Section 4. An application to a real data set is addressed in Section 5. Finally, in Section 6, we offer some concluding remarks.

2. NSINAR(1) process

Let $\mathbb{N}$, $\mathbb{Z}$ and $\mathbb{R}$ denote the set of non-negative integers, integers and real numbers, respectively. All random variables will be defined on the same probability space. In this section, we introduce a stationary first-order integer-valued autoregressive process with geometric–Poisson marginals. With this aim, we first review the new geometric first-order integer-valued autoregressive (NGINAR(1)) process by Ristić et al. [17] and the Poisson INAR(1) process by Al-Osh and Alzaid [1].

The NGINAR(1) process is defined such that

$$X_t = \alpha * X_{t-1} + \epsilon_t, \quad t \in \mathbb{Z},$$

where "$*$" denotes the negative binomial thinning operator [17]. This operator is defined by $\alpha * X \overset{d}{=} \sum_{i=1}^{X} W_i$, where the symbol "$\overset{d}{=}$" means "has the same distribution as" and $\{W_i\}_{i=1}^{\infty}$ is a sequence of independent and identically distributed (i.i.d.) random variables, independent of X, following a geometric distribution with mean $\alpha \in [0, 1)$. Also, $\{\epsilon_t\}_{t\in\mathbb{Z}}$ is a sequence of i.i.d. random variables independent of $\{W_i\}_{i=1}^{\infty}$, with ϵ_t and X_{t-l} being independent for all $l \geq 1$ and distributed such that $\{X_t\}_{t\in\mathbb{Z}}$ is a stationary process having geometric marginals with probability function assuming the form $\Pr(X_t = x) = \mu^x/(1+\mu)^{x+1}$, for all non-negative integers x, where $\mu > 0$.

Ristić et al. [17] proved that the probability mass function of ϵ_t is given by

$$\Pr(\epsilon_t = l) = \left(1 - \frac{\alpha\mu}{\mu - \alpha}\right)\frac{\mu^l}{(1+\mu)^{l+1}} + \frac{\alpha\mu}{\mu - \alpha}\frac{\alpha^l}{(1+\alpha)^{l+1}}, \quad l \in \mathbb{N}, \tag{1}$$

with the condition $\alpha \leq \mu/(1+\mu)$ being necessary to guarantee that all probabilities in (1) are non-negative. The distribution of the random variable ϵ_t is therefore a mixture of two independent geometric distributions with means μ and α. For more details about the NGINAR(1) process, see [17,3,14,15].

The Poisson INAR(1) process introduced by Al-Osh and Alzaid [1] is defined on the discrete support $\mathbb{N}$ of nonnegative integers by means of the difference equation

$$Y_t = \alpha \circ Y_{t-1} + \varepsilon_t, \quad t \in \mathbb{Z}, \tag{2}$$

where the operator "$\circ$" is the binomial thinning operator [21], which is defined by $\alpha \circ Y \stackrel{d}{=} \sum_{i=1}^{Y} U_i$, $\{U_i\}_{i=1}^{\infty}$ being i.i.d. random variables, independent of Y, with $\Pr(U_i = 1) = 1 - \Pr(U_i = 0) = \alpha$. Also, $\{\varepsilon_t\}_{t\in\mathbb{Z}}$ is a sequence of i.i.d. non-negative integer valued random variables which we assume to be Poisson distributed with mean $\lambda(1-\alpha)$, ε_t and Y_{t-i} being independent for all $i \geq 1$ and all t. Thus, $\{Y_t\}_{t\in\mathbb{Z}}$ is a stationary process having Poisson marginals with mean λ. The process $\{Y_t\}_{t\in\mathbb{Z}}$ satisfying (2) is second-order stationary if $0 \leq \alpha < 1$ [5]. For more details about the Poisson INAR(1) process, see [1,7,19,22].

Now, let us consider two random variables X and Y in $\mathbb{N}$ and their difference $Z = X - Y$. The probability function of the difference Z has $\mathbb{Z}$ as support. If X and Y follow independently Geometric(p) and Poisson(λ) distributions with means $\mu = p/(1-p) > 0$ and $\lambda > 0$, respectively, then the random variable $Z = X - Y$ has probability mass function given by

$$\Pr(k;\mu,\lambda) \equiv \Pr(Z=k) = \frac{\mu^k e^{-\lambda/(\mu+1)}}{(1+\mu)^{k+1}} \begin{cases} 1, & k \geq 0, \\ \dfrac{\gamma(-k,\ \lambda\,\mu/(\mu+1))}{\Gamma(-k)}, & k < 0, \end{cases} \tag{3}$$

where $\gamma(a,z) = \frac{1}{\Gamma(a)}\int_0^z t^{a-1}e^{-t}dt$ denotes the incomplete gamma function and $\Gamma(\cdot)$ is the gamma function.

We will denote this distribution as GP(μ,λ). [16] studied the distribution arising from the difference of two discrete random variables belonging to the Panjer family of distributions. They discussed some distributional properties and computation of probabilities. As a special case they defined the difference of the negative binomial and Poisson distributions. The geometric–Poisson distribution is a particular case of this difference.

Suppose that the random variable Z follows a GP(μ,λ) distribution. The moment generating function (mgf) of Z, denoted by $\Phi_Z(s) = \mathrm{E}[\exp(sZ)]$, is given by

$$\Phi_Z(s) = \frac{\exp[\lambda(e^{-s}-1)]}{[1+\mu(1-e^s)]}, \quad s \in \mathbb{R}. \tag{4}$$

It is immediate from (4) that the moments of Z are given by

$$\mathrm{E}(Z^n) = \frac{\partial^n}{\partial s^n}\frac{\exp[\lambda(e^{-s}-1)]}{[1+\mu(1-e^s)]}\bigg|_{s=0}.$$

In particular, the first four moments of Z are

$$\mathrm{E}(Z) = \mu - \lambda, \quad \mathrm{E}(Z^2) = 2\mu^2 + \mu + \lambda^2 + \lambda - 2\mu\lambda,$$
$$\mathrm{E}(Z^3) = 6\mu^3 - 6\lambda\mu^2 + 6\mu^2 + 3\mu\lambda^2 - 3\lambda^2 - \lambda^3 - \lambda + \mu,$$

and

$$\mathrm{E}(Z^4) = \lambda + 7\lambda^2 - 2\lambda\mu + 6\lambda^2\mu + \lambda^4 - 4\lambda^3\mu + 12\lambda^2\mu^2 - 24\lambda\mu^3 + 24\mu^4 + 36\mu^3 + 14\mu^2 + \mu.$$

Further calculations show that the variance, skewness and kurtosis of Z are given by

$$\mathrm{Var}(Z) = \mu(1+\mu) + \lambda, \qquad \gamma_1 = \frac{[\mu(1+\mu)(1+2\mu)-\lambda]}{[\mu(1+\mu)+\lambda]^{3/2}} \quad \text{and}$$

$$\gamma_2 = 3 + \frac{[\mu(1+\mu)(6\mu^2+6\mu+1)+\lambda]}{[\mu(1+\mu)+\lambda]^2},$$

respectively. Thus,

$$\frac{\gamma_2 - 3}{\gamma_1^2} = \frac{[\mu(1+\mu)(6\mu^2+6\mu+1)+\lambda][\mu(1+\mu)+\lambda]}{[\mu(1+\mu)(1+2\mu)-\lambda]^2}.$$

Clearly, $\mathrm{Var}(Z) > |\mathrm{E}(Z)|$ and $\gamma_2 - 3 > \gamma_1^2$. It is also interesting to observe that $\gamma_1 > 0$ when $\lambda = \mu$. For the Skellam (when $\lambda_1 = \lambda_2$) and discrete Laplace distributions (when $\mu_1 = \mu_2$), $\gamma_1 = 0$.

Let Z be a random variable with $\mathrm{GP}(\mu, \lambda)$ distribution. Thus, $Z \stackrel{d}{=} X - Y$, where X and Y are two independent random variables with Geometric$(\mu/(1+\mu))$ and Poisson(λ) distributions, respectively. We define our thinning operator "$\star$" by

$$(\alpha \star Z)|Z \stackrel{d}{=} (\alpha * X - \alpha \circ Y)|(X - Y),$$

where the counting series in $\alpha * X$ and $\alpha \circ Y$ are mutually independent random variables, the counting series in $\alpha * X$ being geometrically distributed with mean α and those in $\alpha \circ Y$ being Bernoulli distributed with mean α.

Now, let X and Y be two independent random variables with Geometric$(\mu/(1+\mu))$ and Poisson(λ) distributions, respectively. It is immediate, from the above definition, that

$$\alpha \star Z \stackrel{d}{=} \alpha * X - \alpha \circ Y.$$

The basic properties of the thinning operator "$\star$" are stated in Proposition 1.

Proposition 1. *Let Z be a random variable with* $\mathrm{GP}(\mu, \lambda)$ *distribution.*

(i) $0 \star Z = 0$.
(ii) $1 \star Z$ *is* not *distributed as* Z.
(iii) $\mathrm{E}(\alpha \star Z) = \alpha(\mu - \lambda)$.
(iv) $\mathrm{Var}(\alpha \star Z) = \alpha\,\mu(1 + 2\,\alpha + \alpha\,\mu) + \alpha\,\lambda$.
If X and Y are independent, X geometrically distributed with mean μ, Y Poisson with mean λ,
(v) $\mathrm{E}(\alpha * X - \alpha \circ Y|X = x, Y = y) = \alpha(x - y)$.
(vi) $\mathrm{Var}(\alpha * X - \alpha \circ Y|X = x, Y = y) = \alpha(1+\alpha)x + \alpha(1-\alpha)y$.
(vii) $\mathrm{E}(\alpha * X - \alpha \circ Y|X - Y = z) = \alpha z$.

Under this operator, the new skew INAR(1) process can be defined.

Definition 2.1 (*NSINAR(1) Process*)**.** Let $\{Z_t\}_{t\in\mathbb{Z}}$ be a sequence of random variables following a common geometric–Poisson distribution with parameters μ and λ, and suppose that ξ_t, for $t \in \mathbb{Z}$, has the distribution of $\epsilon_t - \varepsilon_t$, where the sequences $\{\epsilon_t\}_{t\in\mathbb{Z}}$ and $\{\varepsilon_t\}_{t\in\mathbb{Z}}$ are independent, $\{\epsilon_t\}_{t\in\mathbb{Z}}$ are i.i.d. random variables with common probability function given in (1) and $\{\varepsilon_t\}_{t\in\mathbb{Z}}$ are i.i.d. random variables with common Poisson$(\lambda(1-\alpha))$ distribution. We also suppose that ξ_t and Z_{t-l} are independent for all $l \geq 1$. Under this, we define completely our NSINAR(1) process $\{Z_t\}_{t\in\mathbb{Z}}$ by

$$Z_t = \alpha \star Z_{t-1} + \xi_t, \quad t \in \mathbb{Z}. \tag{5}$$

From the results of Du and Li [5] and Ristić et al. [17] we have that $0 \leq \alpha \leq \mu/(1+\mu)$ and $\alpha \in (\mu/(1+\mu), 1]$ are the conditions of stationarity and non-stationarity of the process $\{Z_t\}_{t\in\mathbb{Z}}$, respectively. Also, $\alpha = 0$ and $\alpha > 0$, respectively, imply the independence and dependence of the observations of $\{Z_t\}_{t\in\mathbb{Z}}$. Here, we restrict our study to the stationary case.

Proposition 2. *The distribution $\alpha \star Z_{t-1}|Z_{t-1}$ is given by*

$$\Pr(\alpha \star Z_{t-1} = j|Z_{t-1} = k) = \frac{\mathrm{e}^{-\frac{\lambda\mu}{1+\mu}}\alpha^j}{(1+\alpha)^{k+j}} \sum_{l=0}^{\infty}\sum_{m=0}^{l} \binom{j+l+k-1}{j}\binom{l}{m}\frac{\alpha^m(1-\alpha)^{l-m}}{(1+\alpha)^l} \times \frac{[\lambda\,\mu/(1+\mu)]^l}{l!},$$

for $k, j \geq 0$.

Remark 3. If $\alpha = 0$, then $Z_t = \xi_t$ for all t and $\{Z_t\}_{t\in\mathbb{Z}}$ is a sequence of i.i.d. random variables having the $GP(\mu, \lambda)$ distribution.

Remark 4. If $\lambda \to 0^+$, $\{Z_t\}_{t\in\mathbb{Z}}$ becomes the NGINAR(1) process proposed by Ristić et al. [17].

3. Properties of the process

In this section we will consider some properties of the NSINAR(1) process, such as the conditional moments, the autocorrelation structure and innovation structure.

Using the definition of ϵ_t and Eq. (1), we obtain that the probability mass function of ξ_t can be expressed by

$$\Pr(\xi_t = k) = \left(1 - \frac{\alpha\mu}{\mu - \alpha}\right) \Pr(k; \mu, \lambda(1-\alpha)) + \frac{\alpha\mu}{\mu - \alpha} \Pr(k; \alpha, \lambda(1-\alpha)),$$

for $k \in \mathbb{Z}$, where $\Pr(k; \cdot, \cdot)$ is defined in (3). Note that the condition $0 \le \alpha \le \mu/(1+\mu)$ assures that the probabilities of the preceding mixture distributions are well defined.

That is, the random variable ξ_t is distributed as a mixture of geometric–Poisson random variables. Using this, it is straightforward to obtain that the mgf of ξ_t, denoted by $\Phi_\xi(s) = \mathrm{E}[\exp(s\,\xi_t)]$, is given by

$$\Phi_\xi(s) = \frac{[1 + \alpha(1+\mu)(1-\mathrm{e}^s)]\exp[\lambda(1-\alpha)(\mathrm{e}^{-s}-1)]}{[1+\mu(1-\mathrm{e}^s)][1+\alpha(1-\mathrm{e}^s)]},$$

for $s \in \mathbb{R}$. We have that the two first cumulants of ξ_t are given by

$$\mathrm{E}(\xi_t) = \mu_\xi = (1-\alpha)(\mu - \lambda),$$
$$\mathrm{Var}(\xi_t) = \sigma_\xi^2 = (1+\alpha)\mu[(1-\alpha)(1+\mu) - \alpha] + \lambda(1-\alpha).$$

Remark 5. If $\alpha = \mu/(1+\mu)$, then ξ_t has a geometric–Poisson distribution with parameters $\mu/(1+\mu)$ and $\lambda/(1+\mu)$.

Proposition 6. *The* NSINAR(1) *process* $\{Z_t\}_{t\in\mathbb{Z}}$ *given by* (5) *is a strict stationary and ergodic process.*

The conditional moment generating function (cmgf) is directly derived using (5) and basic relations of the conditional expectation. So, for $z \ge 0$, we get

$$\mathrm{E}(\mathrm{e}^{sZ_t}|Z_{t-1} = z) = \Phi_\xi(s)[1+\alpha(1-\mathrm{e}^s)]^{-z}\exp\left\{\frac{[\lambda\,\mu/(1+\mu)][(1-\alpha)+\alpha\,\mathrm{e}^{-s}]}{1+\alpha(1-\mathrm{e}^s)} - \frac{\lambda\,\mu}{1+\mu}\right\}.$$

Now, based on the cmgf properties, the conditional expectation and conditional variance of the NSINAR(1) process for given Z_{t-1} are obtained, respectively, as

$$\mathrm{E}(Z_t|Z_{t-1}) = \alpha\,Z_{t-1} + \mu_\xi,$$
$$\mathrm{Var}(Z_t|Z_{t-1}) = \alpha(1+\alpha)|Z_{t-1}| + \frac{2\,\alpha\,\lambda\,\mu}{1+\mu} + \sigma_\xi^2.$$

It is also easy to verify that the autocorrelation function (ACF) at lag h is given by

$$\mathrm{Corr}(Z_t, Z_{t-h}) = \rho(h) = \alpha^h, \quad h \ge 0, \tag{6}$$

for $0 \le \alpha \le \mu/(1+\mu)$. That is, the autocorrelation function decays exponentially as $h \to \infty$.

The spectral density function $f(\omega)$ of any order contains important information about the properties of the process. The spectral density of the NSINAR(1) process is

$$f(\omega) = \frac{\mu(1+\mu)(1-\alpha^2) + \lambda(1-\alpha^2)}{2\pi[1+\alpha^2 - 2\alpha\cos(\omega)]}, \quad \omega \in (-\pi, \pi].$$

4. Estimation and inference of the unknown parameters

This section is concerned with the estimation of the three parameters of interest. For the NSINAR(1), the likelihood estimator is based upon the fact that the conditional distribution $\Pr(Z_t|Z_{t-1})$is the convolution of the difference between the binomial thinning and negative binomial

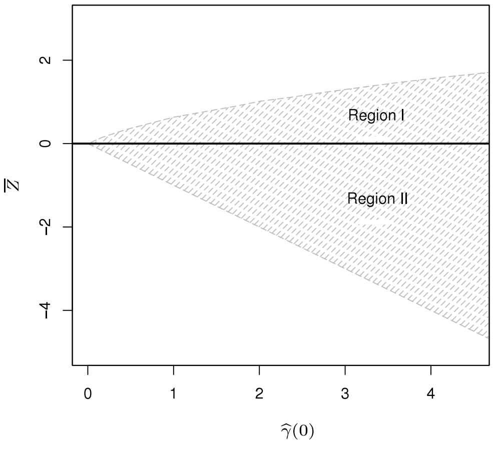

Fig. 1. The plots for the defined regions.

thinning resulting from the random variable $\alpha \star Z_{t-1}$, with the mixture of geometric–Poisson distribution of the innovation ξ_t. Of course, analytical solutions could be obtained by quite tedious calculations (there will be no closed form for the likelihood estimates), but there are a few functions implemented in R programming language, that makes maximum likelihood estimation possible. Since the likelihood for this process is cumbersome to work with, we will use Yule–Walker to estimate the model parameters in this paper.

From a sample $Z_1, \ldots, Z_T$, the Yule–Walker (YW) estimator of α, based upon the fact that $\rho(h) = \alpha^h$, as in (6), is given by

$$\widehat{\alpha} = \frac{\sum_{t=1}^{T-1} (Z_t - \overline{Z})(Z_{t+1} - \overline{Z})}{\sum_{t=1}^{T} (Z_t - \overline{Z})^2},$$

where $\overline{Z} = (1/T) \sum_{t=1}^{T} Z_t$ is the sample mean. We have that the first two cumulants of Z_t are given by $\mathrm{E}(Z_t) = \mu - \lambda$ and $\mathrm{Var}(Z_t) = \mu(1+\mu) + \lambda$. Using this, the moment estimators of μ and λ are defined as

$$\widehat{\mu} = \sqrt{\overline{Z} + \widehat{\gamma}(0) + 1} - 1 \quad \text{and} \quad \widehat{\lambda} = \widehat{\mu} - \overline{Z},$$

respectively, where $\widehat{\gamma}(h) = (1/T) \sum_{t=1}^{T-h} (Z_t - \overline{Z})(Z_{t-h} - \overline{Z})$ is the autocovariance function. Note that the moment estimate for μ is well defined if $\overline{Z} + \widehat{\gamma}(0) > 0$. On the other hand, both moment estimates are well defined for $\overline{Z} > 0$, if $\widehat{\gamma}(0) > \overline{Z}(\overline{Z} + 1)$ (Region I), and for $\overline{Z} < 0$, if $\overline{Z} + \widehat{\gamma}(0) > 0$ (Region II). In simulated samples, cases like this usually happen when μ is very small compared to λ. Fig. 1 displays the plots of these regions for the estimates.

Now, we can derive the asymptotic properties of the YW estimators. They follow from the following proposition, whose proof can be found in the Appendix.

Proposition 7. *The estimator* $(\widehat{\alpha}, \widehat{\mu}, \widehat{\lambda})$ *is strongly consistent and has an asymptotic normal distribution.*

Another idea that is frequently considered in INAR modeling is to estimate the unknown parameters by minimizing

$$\sum_{t=2}^{T} |Z_t - \mathrm{E}[Z_t | Z_{t-1}]|^p,$$

Table 1
Empirical means and mean squared errors (in parentheses) of the estimates of the parameters for $(\mu, \lambda) = (8, 4)$, some values of α and T, and large-T approximation ("$T = \infty$").

T	α	$\widehat{\alpha}$	$\widehat{\mu}$	$\widehat{\lambda}$
100	0.15	0.1332 (0.0097)	7.9002 (1.3816)	3.9156 (0.7485)
	0.30	0.2795 (0.0097)	7.8661 (1.6850)	3.8730 (0.8149)
	0.45	0.4222 (0.0094)	7.8425 (2.0949)	3.8139 (1.0496)
	0.60	0.5615 (0.0092)	7.6840 (2.9914)	3.6977 (1.4583)
	0.75	0.7038 (0.0086)	7.5395 (5.0219)	3.5091 (2.3491)
200	0.15	0.1409 (0.0048)	7.9463 (0.7364)	3.9642 (0.3708)
	0.30	0.2878 (0.0050)	7.9207 (0.8633)	3.9209 (0.4209)
	0.45	0.4360 (0.0045)	7.9192 (1.1075)	3.9229 (0.5629)
	0.60	0.5821 (0.0043)	7.8800 (1.5755)	3.8594 (0.7425)
	0.75	0.7244 (0.0040)	7.6911 (2.6732)	3.7210 (1.2557)
400	0.15	0.1453 (0.0025)	7.9705 (0.3533)	3.9790 (0.1845)
	0.30	0.2941 (0.0024)	7.9724 (0.4241)	3.9723 (0.2118)
	0.45	0.4419 (0.0023)	7.9458 (0.5512)	3.9426 (0.2697)
	0.60	0.5909 (0.0022)	7.9228 (0.8386)	3.9164 (0.4027)
	0.75	0.7374 (0.0019)	7.8387 (1.4162)	3.8559 (0.6256)
800	0.15	0.1473 (0.0013)	7.9891 (0.1910)	3.9873 (0.0986)
	0.30	0.2967 (0.0013)	7.9750 (0.2238)	3.9806 (0.1116)
	0.45	0.4464 (0.0012)	7.9736 (0.2756)	3.9658 (0.1372)
	0.60	0.5956 (0.0010)	7.9581 (0.4003)	3.9549 (0.1927)
	0.75	0.7435 (0.0009)	7.9512 (0.7465)	3.9162 (0.3271)
∞	0.15	0.15 (9.977×10^{-6})	8.0 (1.485×10^{-3})	4.0 (7.409×10^{-4})
	0.30	0.40 (9.998×10^{-6})	8.0 (1.770×10^{-3})	4.0 (8.912×10^{-4})
	0.45	0.45 (9.209×10^{-6})	8.0 (2.256×10^{-3})	4.0 (1.106×10^{-3})
	0.60	0.60 (8.213×10^{-6})	8.0 (3.312×10^{-3})	4.0 (1.577×10^{-3})
	0.75	0.75 (7.314×10^{-6})	8.0 (5.876×10^{-3})	4.0 (2.748×10^{-3})

for a given $p \geq 1$. For example, the case $p = 2$ will give us conditional least squares estimation. However, in our case, $\mathrm{E}[Z_t|Z_{t-1}] = \alpha Z_{t-1} + (1-\alpha)(\mu - \lambda)$, which depends on μ and λ only through $\mu - \lambda$. Therefore, we cannot define estimators of μ and λ with this technique, in our case. For this reason, this kind of estimation was not were considered.

Next, a small Monte Carlo simulation experiment will be conducted to evaluate the estimation of the NSINAR(1) process parameters. The simulation was performed using the R programming language; see http://www.r-project.org. The number of Monte Carlo replications was $R = 5000$. The sample sizes considered are $T = 100, 200, 400, 800$ and the values of the parameters are the combinations $\{(\mu, \lambda) = (8, 4), (\mu, \lambda) = (4, 8)\} \times \{\alpha = 0.15, \alpha = 0.30, \alpha = 0.45, \alpha = 0.60, \alpha = 0.75\}$. Furthermore, as suggested by one of the referees, we computed large-T approximations (blocks "$T = \infty$" in Tables 1 and 2).

Tables 1 and 2 present the empirical mean and mean squared error of the estimates of the parameters of the NSINAR(1) process. From those tables, notice that the biases and the mean square errors decrease as the size of the sample increases, evidencing that the estimators are asymptotically unbiased.

5. Application to the Saudi stock market data

In this section we present an application of the NSINAR(1) model to the Saudi stock market. We consider the daily difference between the close and open prices in 2012 of the Saudi Telecom. Since these differences belong to $\mathbb{Z}/10$, we work with the rescaled time series $Z =$ (close price − open price) $\times$ 10, which we call number of ticks. The number of ticks belongs to $\mathbb{Z}$ and is our time series of interest here in this section. The data consists of 251 observations. The data has been downloaded from the Saudi Stock Exchange (http://www.tadawul.com.sa).

Table 3 provides some descriptive measures for the number of ticks, including central tendency statistics, variance, skewness and kurtosis, variance of Z^+ and variance of Z^-, where Z^+ and Z^-

Table 2
Empirical means and mean squared errors (in parentheses) of the estimates of the parameters for $(\mu, \lambda) = (4, 8)$, some values of α and T, and large-T approximation ("$T = \infty$").

T	α	$\widehat{\alpha}$	$\widehat{\mu}$	$\widehat{\lambda}$
100	0.15	0.1327 (0.0098)	3.9273 (0.4559)	7.9270 (0.3355)
	0.30	0.2758 (0.0100)	3.9364 (0.5592)	7.9157 (0.4172)
	0.45	0.4225 (0.0098)	3.8875 (0.6874)	7.8787 (0.5400)
	0.60	0.5638 (0.0092)	3.8163 (1.0493)	7.8069 (0.7827)
	0.75	0.6995 (0.0094)	3.6673 (1.7430)	7.6685 (1.2797)
200	0.15	0.1438 (0.0051)	3.9814 (0.2281)	7.9753 (0.1738)
	0.30	0.2880 (0.0049)	3.9653 (0.2793)	7.9706 (0.2171)
	0.45	0.4347 (0.0048)	3.9390 (0.3479)	7.9343 (0.2772)
	0.60	0.5806 (0.0045)	3.9016 (0.5462)	7.9095 (0.3942)
	0.75	0.7241 (0.0040)	3.8203 (0.9354)	7.8166 (0.6349)
400	0.15	0.1463 (0.0025)	3.9885 (0.1167)	7.9908 (0.0867)
	0.30	0.2942 (0.0025)	3.9875 (0.1409)	7.9805 (0.1069)
	0.45	0.4443 (0.0024)	3.9798 (0.1822)	7.9746 (0.1364)
	0.60	0.5908 (0.0021)	3.9509 (0.2612)	7.9563 (0.2027)
	0.75	0.7359 (0.0019)	3.8994 (0.4835)	7.9043 (0.3407)
800	0.15	0.1480 (0.0012)	3.9928 (0.0571)	7.9941 (0.0445)
	0.30	0.2977 (0.0012)	3.9884 (0.0682)	7.9900 (0.0533)
	0.45	0.4465 (0.0012)	3.9853 (0.0914)	7.9848 (0.0700)
	0.60	0.5949 (0.0011)	3.9680 (0.1435)	7.9683 (0.1013)
	0.75	0.7426 (0.0009)	3.9473 (0.2473)	7.9498 (0.1778)
∞	0.15	0.15 (1.037×10^{-5})	4.0 (4.555×10^{-4})	8.0 (3.551×10^{-4})
	0.30	0.30 (9.851×10^{-6})	4.0 (5.789×10^{-4})	8.0 (4.330×10^{-4})
	0.45	0.45 (9.533×10^{-6})	4.0 (7.274×10^{-4})	8.0 (5.824×10^{-4})
	0.60	0.60 (8.399×10^{-6})	4.0 (1.144×10^{-3})	8.0 (8.195×10^{-4})
	0.75	0.75 (7.264×10^{-6})	4.0 (2.039×10^{-3})	8.0 (1.334×10^{-3})

Table 3
Descriptive statistics.

Min.	Median	Mean	Var(Z)	Var(Z^+)	Var(Z^-)	Skewness	Kurtosis	$\widehat{\rho}(1)$	Max.
−18	0	0.5259	18.066	13.346	6.1147	0.9337	8.1169	0.1795	22

are the positive and the negative parts of Z, respectively: $Z^+ = \max(0, Z)$ and $Z^- = (-Z)^+$. In Fig. 2, we have the histogram for the number of ticks. We see that the data set assumes positive and negative integer values. These measures and the histogram indicate the positively skewed nature of the distribution underlying these data. Furthermore, the sample variance of Z^+ is much larger than the sample variance of Z^-.

The time series data and their sample autocorrelation and partial autocorrelation are displayed in Fig. 3. Analyzing Fig. 3, we conclude that a first order autoregressive model may be appropriate for the given data series, because of the clear cut-off after lag 1 in the partial autocorrelations. Furthermore, the behavior of the series indicates that it may be mean stationary.

We use the estimators of Section 4 to fit a NSINAR(1) model with $(\widehat{\alpha}, \widehat{\mu}, \widehat{\lambda}) = (0.1795, 3.4263, 2.9004)$. The standard errors for the estimates of the parameters $\widehat{\alpha}$, $\widehat{\mu}$ and $\widehat{\lambda}$ are respectively 0.0627, 0.3507 and 0.2743; the standard errors of $\widehat{\alpha}$, $\widehat{\mu}$ and $\widehat{\lambda}$ are obtained via Monte Carlo simulation.

In order to check the adequacy of our fitted model, we plot the sample ACF of the residuals and the jumps against time, which is defined by $J_t \equiv Z_t - Z_{t-1}$, (this plot is known as the Shewhart control chart) with $\pm 3\sigma_J$ as the limits chosen as the benchmark chart. We here define $\sigma_J = \sqrt{\mathrm{Var}(J_t)}$, where the variance of J_t is given by $2(1-\alpha)(\lambda + \mu + \mu^2)$. These plots are presented in Fig. 4. They indicate that the residuals are not correlated and that there is no particular point causing a huge impact in the model. With this, our NSINAR(1) model seems to be well fitted to the number of ticks of the Saudi Telecom in 2012.

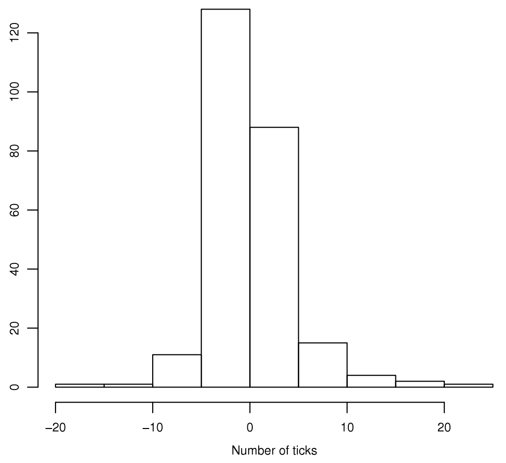

Fig. 2. Histogram for the number of ticks of the Saudi Telecom in 2012.

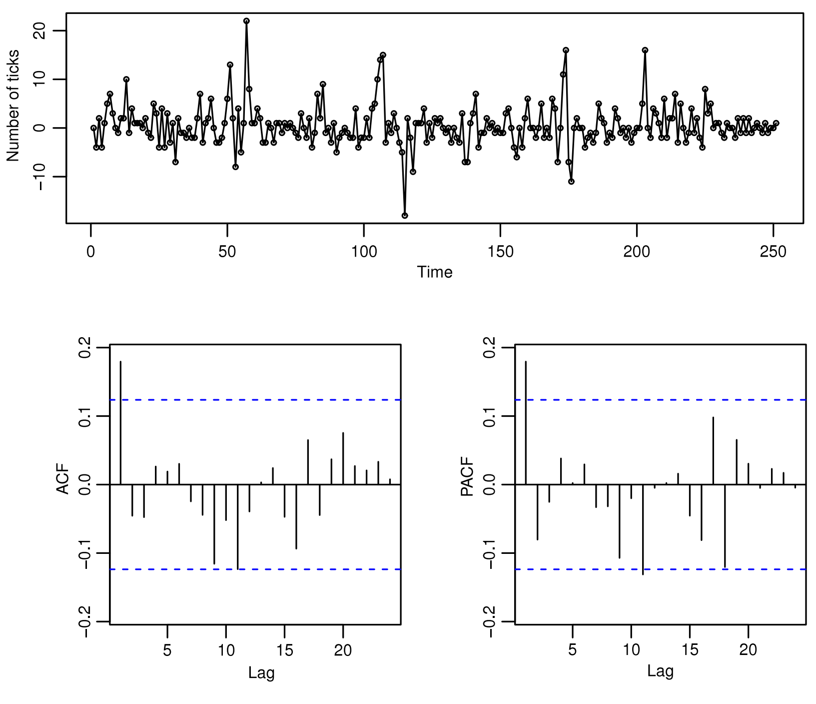

Fig. 3. Time series plot, autocorrelation and partial autocorrelation functions for the number of ticks of the Saudi Telecom in 2012.

6. Concluding remarks

In this paper, we introduce a stationary first-order integer-valued autoregressive process with geometric–Poisson marginals. The new model has several advantages: possible negative values for time series and simple innovation structure. The main properties of the model are derived. Then we considered the problem of parameter estimation and derived the asymptotic normality distribution of the Yule–Walker estimators. An example of application to a real data set illustrates the importance

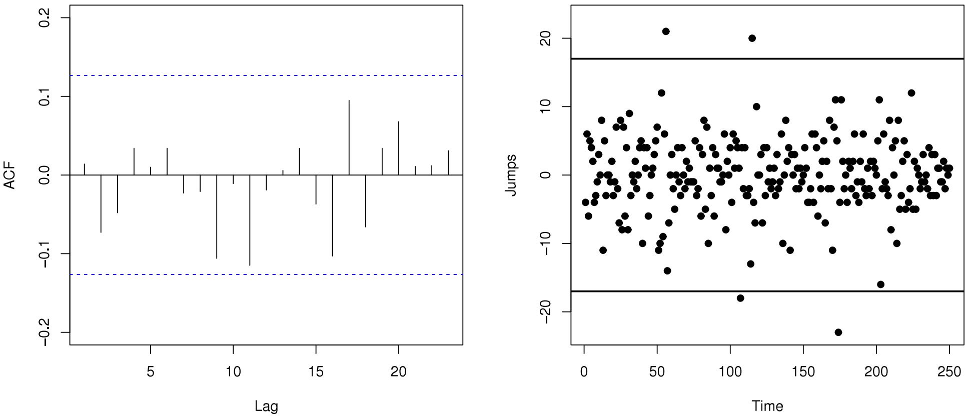

Fig. 4. Sample autocorrelation function of the residuals and jumps against time of our fitted SINARZ(1) model.

and potentiality of the new model. We hope this new process may attract wider applications in count time series analysis. As part of future research, it would be of interest to extend the proposed process, studying the process of order p.

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Appendix

Proof of Proposition 1. (i) is trivial.

(ii) follows from the fact that $1 \star Z$ is distributed as $1 * X - Y$, where X is geometrically distributed with mean μ, Y is Poisson(λ), X and Y independent. Then it suffices to verify that $1*X$ is not distributed as X, which is easy to check. For example, $\Pr(1*X = 0) = \mathrm{E}[\Pr(1*X = 0|X)] = \mathrm{E}[2^{-X}] = 2/(2+\mu)$, while $\Pr(X = 0) = 1/(1+\mu)$.

(iii) and (iv) follow from the fact that $\alpha \star Z$ is distributed as $\alpha * X - \alpha \circ Y$, where X is geometrically distributed with mean μ, Y is Poisson(λ), X and Y independent.

(v) and (vi) will follow easily, since X and Y are independent.

(vii) can be deduced from the following general fact: let V be a discrete random variable such that $E|V| < \infty$ and let $\{A_k\}_{k\in\mathbb{Z}}$ be a sequence of random events with $\Pr(A_k) > 0$ for all k. Suppose that $\mathrm{E}[V|A_k]$ is a constant c not depending on k. Then, $\mathrm{E}[V| \cup A_k]$ is also equal to c.

Proof of Proposition 2. We have

$$\begin{aligned}
&\Pr(\alpha \star Z_{t-1} = j|Z_{t-1} = k)\\
&\quad= \frac{\Pr(\alpha \star Z_{t-1} = j, Z_{t-1} = k)}{\Pr(Z_{t-1} = k)}\\
&\quad= \frac{1}{\Pr(Z_{t-1} = k)} \sum_{l=0}^{\infty} \Pr(\alpha \star Z_{t-1} = j, X_{t-1} = k + l, Y_{t-1} = l)\\
&\quad= \frac{1}{\Pr(Z_{t-1} = k)} \sum_{l=0}^{\infty} \Pr(\alpha \star Z_{t-1} = j|X_{t-1} = k + l, Y_{t-1} = l)\Pr(X_{t-1} = k + l, Y_{t-1} = l)\\
&\quad= \mathrm{e}^{-\frac{\lambda\mu}{1+\mu}} \sum_{l=0}^{\infty} \frac{[\lambda\,\mu/(1+\mu)]^l}{l!} \Pr(\alpha \star Z_{t-1} = j|X_{t-1} = k + l, Y_{t-1} = l). \qquad (7)
\end{aligned}$$

We know that the random variables $\alpha * X_{t-1}|X_{t-1} = k+l$ and $\alpha \circ Y_{t-1}|Y_{t-1} = l$ have $\mathrm{NB}(k+l, \alpha/(1+\alpha))$ and $\mathrm{B}(l,\alpha)$ distributions. Thus,

$$\Pr(\alpha \star Z_{t-1} = j|X_{t-1} = k+l, Y_{t-1} = l) = \sum_{m=0}^{l} \binom{j+l+k-1}{j}\binom{l}{m}\frac{\alpha^{j+m}(1-\alpha)^{l-m}}{(1+\alpha)^{l+k+j}} \tag{8}$$

plugging (8) into (7), we obtain

$$\Pr(\alpha \star Z_{t-1} = j|Z_{t-1} = k) = \frac{e^{-\frac{\lambda\mu}{1+\mu}}\alpha^j}{(1+\alpha)^{k+j}}\sum_{l=0}^{\infty}\sum_{m=0}^{l}\binom{j+l+k-1}{j}\binom{l}{m}\frac{\alpha^m(1-\alpha)^{l-m}}{(1+\alpha)^l} \times \frac{[\lambda\mu/(1+\mu)]^l}{l!}.$$

Proof of Proposition 6. The strict stationarity of the NSTINAR(1) process $\{Z_t\}_{t\in\mathbb{Z}}$ follows from the fact that it is a Markov process of first order and that the random variables $\{Z_t\}_{t\in\mathbb{Z}}$ are identically distributed random variables. Now, we will prove ergodicity. Let $\{V(t)\}$ be all counting series of $\alpha\star Z_{t-1}$ in (5). Let $\sigma(U_1, U_2, \ldots)$ denote the σ-field generated by the sequence of random variables $U_1, U_2, \ldots$; from (5), we obtain

$$\sigma(Z_t, Z_{t-1}, \ldots) \subset \sigma(\xi_t, V(t), \xi_{t-1}, V(t-1), \ldots)$$

and hence

$$\bigcap_{t=0}^{-\infty}\sigma(Z_t, Z_{t-1}, \ldots) \subset \bigcap_{t=0}^{-\infty}\sigma(\xi_t, V(t), \xi_{t-1}, V(t-1), \ldots). \tag{9}$$

Since $\{V(t), \xi_t\}$ are independent sequences, then, from Kolmogorov's Zero–One law, any event in the tail σ-field denoted by (9) has probability 0 or 1. From Lemma 2 of [18, pp. 408], $\{Z_t\}_{t\in\mathbb{Z}}$ is ergodic. For more details, see [15].

Proof of Proposition 7. Strong consistency of these estimators follows from ergodicity of the process. For the NSTINAR(1) process, we have that $\sum_{i=0}^{\infty}\alpha^i < \infty$. Thus, the sufficient condition of Theorem 1 of [20] is satisfied for the NSTINAR(1) process. So, we obtain that the estimator $(\overline{Z}, \widehat{\gamma}(0), \widehat{\gamma}(1))^\top$ has asymptotic normal distribution with mean $(\mu-\lambda, \mu(1+\mu)+\lambda, \alpha[\mu(1+\mu)+\lambda])^\top$ and variance $T^{-1}\boldsymbol{\Sigma}$, where the matrix $\boldsymbol{\Sigma}$ is defined as in Theorem 1 of [20].

Now, we use the delta method: suppose $\boldsymbol{X} = (x_1, x_2, x_3)^\top$, $\boldsymbol{\eta}$ and $\boldsymbol{\Sigma}$ are such that $\sqrt{T}(\boldsymbol{X}-\boldsymbol{\eta}) \xrightarrow{d} \mathrm{N}(0, \boldsymbol{\Sigma})$. Let $f = (f_1, f_2, f_3)^\top$ be a mapping from $\mathbb{R}^3$ to $\mathbb{R}^3$ where each f_i is differentiable at $\boldsymbol{\eta}$. Let $\mathbf{D}$ be the Jacobian matrix of f with respect to $\boldsymbol{X}$. Then, $\sqrt{T}(f(\boldsymbol{X}) - f(\boldsymbol{\eta})) \xrightarrow{d} \mathrm{N}\left(0, \mathbf{D}\boldsymbol{\Sigma}\mathbf{D}^\top\right)$.

For $f_1(x_1,x_2,x_3) = x_3/x_2$, $f_2(x_1,x_2,x_3) = \sqrt{x_2+x_1+1}-1$, $f_3(x_1,x_2,x_3) = \sqrt{x_2+x_1+1}-x_1-1$ and $\boldsymbol{\eta} = (\mu-\lambda, \mu(1+\mu)+\lambda, \alpha[\mu(1+\mu)+\lambda])^\top$, we obtain

$$\frac{\partial f_1(x_1,x_2,x_3)}{\partial x_1} = 0, \quad \frac{\partial f_1(x_1,x_2,x_3)}{\partial x_2} = -\frac{x_3}{x_2^2}, \quad \frac{\partial f_1(x_1,x_2,x_3)}{\partial x_3} = \frac{1}{x_2},$$
$$\frac{\partial f_2(x_1,x_2,x_3)}{\partial x_1} = \frac{1}{\sqrt{x_1+x_2+1}}, \quad \frac{\partial f_2(x_1,x_2,x_3)}{\partial x_2} = \frac{1}{\sqrt{x_1+x_2}}, \quad \frac{\partial f_2(x_1,x_2,x_3)}{\partial x_3} = 0,$$
$$\frac{\partial f_3(x_1,x_2,x_3)}{\partial x_1} = \frac{1}{\sqrt{x_1+x_2+1}} - 1,$$
$$\frac{\partial f_3(x_1,x_2,x_3)}{\partial x_2} = \frac{1}{\sqrt{x_1+x_2+1}}, \quad \frac{\partial f_3(x_1,x_2,x_3)}{\partial x_3} = 0.$$

Thus, $\sqrt{T}\left(f\left(\overline{Z}, \widehat{\gamma}(0), \widehat{\gamma}(1)\right) - (\alpha, \mu, \lambda)^{\top}\right) \xrightarrow{d} \mathrm{N}\left(0, \mathbf{D}\boldsymbol{\Sigma}\mathbf{D}^{\top}\right)$, with

$$\mathbf{D} = \begin{pmatrix} 0 & -\dfrac{\alpha}{\mu(1+\mu)+\lambda} & \dfrac{1}{\mu(1+\mu)+\lambda} \\ \dfrac{1}{1+\mu} & \dfrac{1}{1+\mu} & 0 \\ -\dfrac{\mu}{1+\mu} & \dfrac{1}{1+\mu} & 0 \end{pmatrix}.$$

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